# Akoizo Original Practice - Form 4 - Answer key

### Division B | Experimental Design | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

### ANSWER KEY - Review only after submitting.

**1. (2 points) Sample SD for 8,10,12 is…**

A. 10

B. 8

C. 2

D. 4

**Correct: C**

**2. (2 points) A pilot primarily checks…**

Unit: The experiment and the prompt packet.
Worked-case data: A prompt supplies a ramp, object, ruler, and timer. Turn “study motion” into a testable question.

A. Whether to delete awkward results

B. The final conclusion automatically

C. Every possible mechanism

D. Feasibility and procedure

**Correct: D**

**3. (2 points) Which addresses time-order drift?**

Unit: Errors, conclusions, and recommendations.
Worked-case data: Every timer reading is 0.5 s too high. Would ten more repeats eliminate the bias?

A. Deleting late trials silently

B. Randomization or suitable blocking

C. Only more decimals

D. Only a larger title

**Correct: B**

**4. (2 points) A reproducible timing method needs…**

Unit: Materials, reproducible procedures, and qualitative data.
Worked-case data: A trial has no timer reading because the observer missed the endpoint. What should the table and procedure record?

A. Only a final mean

B. Only a title

C. Explicit start and stop events

D. Only a stopwatch brand

**Correct: C**

**5. (2 points) A useful practice reflection leads to…**

Unit: Practice, timing, teamwork, and final review.
Worked-case data: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

A. A specific correction tested on a new task

B. Only repeating the same answer

C. Blaming all variation on humans

D. Removing the error section

**Correct: A**

**6. (4 points) Write the example's claim and evidence.**

Unit: Claims, evidence, and reasoning for trends and outliers.
Worked-case data: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.

answer: Mean response increases over levels 1-3; means are 12,16,20 with SD 2, an overall rise of 8 units. (4 points).

**7. (4 points) Distinguish control condition and controlled variable.**

Unit: Problem, hypothesis, variables, and experimentation.
Worked-case data: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

answer: A control condition is a reference treatment; a controlled variable is held comparable across treatments. (4 points).

**8. (4 points) How can roles create contradictions?**

Unit: Practice, timing, teamwork, and final review.
Worked-case data: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

answer: If teammates use different units, endpoints, variable names, or procedure versions without coordination. (4 points).

**9. (4 points) Why show with/without-outlier results?**

Unit: Claims, evidence, and reasoning for trends and outliers.
Worked-case data: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.

answer: It reveals whether the conclusion depends strongly on the flagged value under transparent alternatives. (4 points).

**10. (4 points) Why does replication not solve drift alone?**

Unit: Problem, hypothesis, variables, and experimentation.
Worked-case data: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

answer: More trials can reproduce the same systematic level-time distortion if order remains confounded. (4 points).

**11. (2 points) Define “Range” in the context of Experimental Design.**

answer: Largest value minus smallest value. (2 points).

**12. (6 points) Solve this case and explain your reasoning: For response values 8, 10, 12, calculate mean, range, and sample SD.**

reasoning: Mean = 30/3 = 10. Range = 12 - 8 = 4.
Squared deviations are 4, 0, 4; sum = 8.
Sample variance = 8/(3-1) = 4; sample SD = 2. (3 points).

conclusion: Mean 10, range 4, and sample SD 2 have distinct meanings; variance is 4 squared response units. (3 points).

**13. (2 points) Define “Independent variable” in the context of Experimental Design.**

answer: Factor deliberately changed. (2 points).

**14. (2 points) Define “Conclusion” in the context of Experimental Design.**

answer: A claim supported by the recorded evidence. (2 points).

**15. (2 points) Define “Sample standard deviation” in the context of Experimental Design.**

answer: Spread estimate using squared deviations and n - 1. (2 points).

**16. (2 points) Define “Replication” in the context of Experimental Design.**

answer: Repeated independent experimental units or trials as specified. (2 points).

**17. (2 points) Define “Guard digits” in the context of Experimental Design.**

answer: Extra intermediate digits retained to avoid rounding error. (2 points).

**18. (2 points) Define “Extrapolation” in the context of Experimental Design.**

answer: Extending a relationship beyond observed conditions. (2 points).